

1. Programmers A and programmer B are working on a software project. The number of lines of code they write per day for 6 days as follow:

$$\text{Programmer A} = [50, 60, 45, 55, 70, 65]$$

$$\text{Programmer B} = [45, 55, 50, 60, 65, 70]$$

(i) Calculate the person correlation coefficient between the productivity of programmer A and programmer B based on the number of lines of code they write each day and comment on your result.

(ii) Draw a scatter diagram to represent the above information.

Soln:

(i) Given,

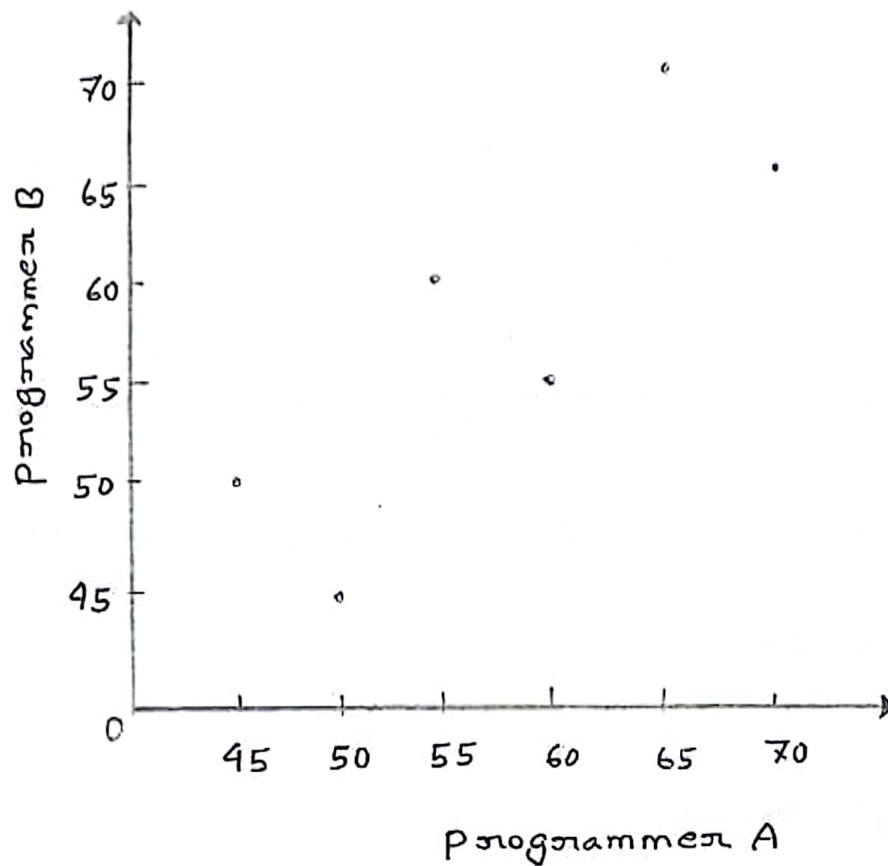
x	y	x^2	y^2	xy
50	45	2500	2025	2250
60	55	3600	3025	3300
45	50	2025	2500	2250
55	60	3025	3600	3300
70	65	4900	4225	4550
65	70	4225	4900	4550
$\Sigma x = 345$	$\Sigma y = 345$	$\Sigma x^2 = 20275$	$\Sigma y^2 = 20275$	$\Sigma xy = 20200$

We know,

$$\begin{aligned} r_{xy} &= \frac{n \Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{\{n \Sigma x^2 - (\Sigma x)^2\} \{n \Sigma y^2 - (\Sigma y)^2\}}} \\ &= \frac{6 \times 20200 - 345 \times 345}{\sqrt{\{6 \times 20275 - (345)^2\} \{6 \times 20275 - (345)^2\}}} \\ &= 0.8286 \end{aligned}$$

Comment: A Pearson correlation coefficient of 0.83 indicates a strong positive co-relation between programmer A and B's productivity.

(ii) Scatter diagram: Programmer A Vs Programmer B.



2. You are analyze the performance of a computer algorithm. You have collected data on the input size (n) and the corresponding execution time (y) for different input size:

$$n : [100, 120, 140, 150, 160]$$

$$y : [12, 22, 28, 31, 40]$$

- (i) Determine the regression line of y on x for this data.
- (ii) Estimate the execution time for an input size of 180.
- (iii) Compute the regression line of x on y .
- (iv) predict the input size for an execution time of 45.

Solⁿ:- Given,

x	y	x^2	y^2	xy
100	12	10000	144	1200
120	22	14400	484	2640
140	28	19600	784	3920
150	31	22500	961	4650
160	40	25600	1600	6400
$\Sigma x = 670$	$\Sigma y = 133$	$\Sigma x^2 = 92100$	$\Sigma y^2 = 3973$	$\Sigma xy = 18810$

$$\begin{aligned} \text{Hence, } \bar{x} &= \frac{670}{5} & \bar{y} &= \frac{133}{5} \\ &= 134 & &= 26.6 \end{aligned}$$

(i) We know, $y = a + bx$

$$\begin{aligned} \therefore b &= \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} \\ &= \frac{5 \times 18810 - 670 \times 133}{5 \times 2100 - 670^2} \\ &= 0.4259 \end{aligned}$$

$$\begin{aligned} \text{Now, } a &= \bar{y} - b\bar{x} \\ &= 26.6 - 0.4259 \times 134 \\ &= -30.47 \end{aligned}$$

$\therefore$ Regression line of y on x :

$$y = -30.47 + 0.4259x$$

[Ans.]

(ii) Estimate the execution time for input size 180.

By using regression line :

$$\hat{y} = -30.47 + 0.4259 \times 180$$

$$\therefore \hat{y} = 46.15$$

[Ans.]

(iii) We know,

$$m = a' + b'y$$

$$\begin{aligned}\therefore b' &= \frac{n \sum my - \sum m \sum y}{n \sum y^2 - (\sum y)^2} \\ &= \frac{5 \times 18810 - 670 \times 133}{5 \times 3973 - 133^2} \\ &= 2.269\end{aligned}$$

$$\text{Now, } a' = \bar{m} - b'\bar{y}$$

$$= 134 - 2.269 \times 26.6$$

$$= 73.64$$

∴ Regression line of x on y :

$$x = 73.64 + 2.269y$$

[Ans.]

(iv) Predict input size for execution time of 45.

$$x = 73.64 + 2.269 \times 45$$

$$\therefore x = 175.75$$

[Ans.]

3. State Bayes theorem, Three routers - R1, R2

and R3 - forward 40%, 35% and 25% of the total data package respectively. The probability of packet loss for R1 is 0.5%, for R2 is 1% and for R3 is 1.5%. If a packet is found to be lost, find the probability that it was routed through:

(i) R2 ?

(ii) R3 ?

Solⁿ:

Statement of Bayes theorem: Let, $B_1, B_2, \dots, B_n$ be n mutually exclusive and exhaustive events in a random experiment and D be any event in S . Then Bayes theorem states that,

$$P[B_i | D] = \frac{P[B_i] \cdot P[D | B_i]}{\sum_{j=1}^n P[B_j] \cdot P[D | B_j]} ; i = 1, 2, 3, \dots, n$$

We have, packet forwarding probabilities:

$$P(R_1) = 0.40$$

$$P(R_2) = 0.35$$

$$P(R_3) = 0.25$$

Packet loss probability:

$$P(L|R_1) = 0.005$$

$$P(L|R_2) = 0.01$$

$$P(L|R_3) = 0.015$$

(i) According to Bayes theorem,

$$\begin{aligned} P(R_2|L) &= \frac{P(R_2) \cdot P(L|R_2)}{P(R_1) \cdot P(L|R_1) + P(R_2) \cdot P(L|R_2) + P(R_3) \cdot P(L|R_3)} \\ &= \frac{0.35 \times 0.01}{(0.40 \times 0.005) + (0.35 \times 0.01) + (0.25 \times 0.015)} \\ &= 0.378 \end{aligned}$$

$\therefore$ Probability that a lost packet came through R_2 is 0.378.

(ii) According to Bayes theorem,

$$\begin{aligned}P(R_3|L) &= \frac{P(R_3) \cdot P(L|R_3)}{P(R_1) P(L|R_1) + P(R_2) P(L|R_2) + P(R_3) P(L|R_3)} \\&= \frac{0.25 \times 0.015}{(0.40 \times 0.005) + (0.35 \times 0.01) + (0.25 \times 0.015)} \\&= 0.405\end{aligned}$$

$\therefore$ Probability that a lost packet came through

R_3 is 0.405.

4. A company has three servers - Server A,

Server B, Server C - handling 30%, 45% and

25% of the total request respectively. Server

A fails 1% of the time, Server B fails 2% and

Server C fails 3%. A request is found to

have failed. What is the probability that it

was handled by:

(i) Server A ? (ii) Server C ?

Solⁿ: We have,

Handling probability:

$$P(A) = 0.30$$

$$P(B) = 0.45$$

$$P(C) = 0.25$$

Failure probability:

$$P(F|A) = 0.01$$

$$P(F|B) = 0.02$$

$$P(F|C) = 0.03$$

(i) According to Bayes theorem,

$$\begin{aligned} P(F|A) &= \frac{P(A) \cdot P(F|A)}{P(A) \cdot P(F|A) + P(B) \cdot P(F|B) + P(C) \cdot P(F|C)} \\ &= \frac{0.30 \times 0.01}{0.30 \times 0.01 + 0.45 \times 0.02 + 0.25 \times 0.03} \\ &= 0.154 \end{aligned}$$

$\therefore$ Probability it was handled by server A:

$$0.154.$$

(ii) According to Bayes theorem,

$$\begin{aligned}P(C|F) &= \frac{P(C) P(F|C)}{P(A) P(F|A) + P(B) P(F|B) + P(C) P(F|C)} \\&= \frac{0.25 \times 0.03}{0.30 \times 0.01 + 0.40 \times 0.02 + 0.25 \times 0.03} \\&= 0.385\end{aligned}$$

$\therefore$ Probability it was handled by server C is 0.385.

5. A continuous random variable x has the following probability density function:

$$f(x) = \frac{1}{24} (x^2 + 1) ; 1 < x < 4$$

Compute (i) the value of $E(x)$

(ii) $SD(x)$

(iii) $P(x < 3.5)$

(iv) $P(x > 0.3)$

Solⁿ:- Given that,

$$f(u) = \frac{1}{24} (u^2 + 1); \quad 1 < u < 4$$

(i) We know,

$$\begin{aligned} E(x) &= \int_1^4 u f(u) du \\ &= \int_1^4 u \cdot \frac{1}{24} (u^2 + 1) du \\ &= \int_1^4 \frac{1}{24} \cdot (u^3 + u) du \\ &= \frac{1}{24} \left[\frac{u^4}{4} + \frac{u^2}{2} \right]_1^4 \\ &= \frac{1}{24} \left[\left(\frac{4^4}{4} + \frac{4^2}{2} \right) - \left(\frac{1^4}{4} + \frac{1^2}{2} \right) \right] \\ &= \frac{1}{24} (72 - 0.75) \\ &= 2.9687 \quad [Ans.] \end{aligned}$$

(ii) We know,

$$\begin{aligned} E(x^2) &= \int_1^4 u^2 \cdot \frac{1}{24} (u^2 + 1) du \\ &= \frac{1}{24} \int_1^4 (u^4 + u^2) du \end{aligned}$$

$$= \frac{1}{24} \left[\frac{x^5}{5} + \frac{x^3}{3} \right]_1^4$$

$$= \frac{1}{24} \left[\left(\frac{4^5}{5} + \frac{4^3}{3} \right) - \left(\frac{1^5}{5} + \frac{1^3}{3} \right) \right]$$

$$= 0.4$$

Now, $V_{\text{var}}(X) = E(X^2) - [E(X)]^2$

$$= 0.4 - (2.06875)^2$$

$$= 0.586$$

$$\therefore SD(X) = \sqrt{V_{\text{var}}(X)}$$

$$= \sqrt{0.586} = 0.765 \quad [A_m.]$$

(iii) $P(X < 3.5) = \int_1^{3.5} f(x) dx$

$$= \int_1^{3.5} \frac{1}{24} (x^2 + 1) dx$$

$$= \frac{1}{24} \left[\frac{x^3}{3} + x \right]_1^{3.5}$$

$$= \frac{1}{24} \left[\left(\frac{3.5^3}{3} + 3.5 \right) - \left(\frac{1^3}{3} + 1 \right) \right] = 0.686 \quad [A_m.]$$

(iv) Compute $p(u > 0.3)$.

But the limitation of u is $1 < u < 4$.

That's why,

$$= \int_1^4 \frac{1}{24} (u+1) du$$

$$= \frac{1}{24} \int_1^4 (u+1) du$$

$$= \frac{1}{24} \left[\frac{u^2}{2} + u \right]_1^4$$

$$= \frac{1}{24} \left[\left(\frac{16}{2} + 4 \right) - \left(\frac{1}{2} + 1 \right) \right]$$

$$= 1$$

[Ans.]

6. A continuous random value X has the

following probability density function:

$$f(u) = k(u-1); \quad 2 \leq u \leq 6$$

Compute (i) The value of k

(ii) $E(u)$

(iii) $CV(u)$

Solⁿ:- Given that,

$$f(u) = \frac{3}{12} (6u - 3u^2) ; 0 < u < 2$$

(i) The probability that at a given day in this region the rainfall is not more than 1.5 inches -

$$\therefore P(u \leq 1.5) = \int_0^{1.5} \frac{3}{12} (6u - 3u^2) du$$

$$= \frac{1}{4} [3u^2 - u^3]_0^{1.5}$$

$$= \frac{1}{4} \{ 3 \cdot (1.5)^2 - (1.5)^3 \}$$

$$= 0.84375 \quad [Ans.]$$

(ii) The probability that at a given day in this region the rainfall is between 0.5 and 1.5 inches -

$$\therefore P(0.5 < u < 1.5) = \int_{0.5}^{1.5} f(u) du$$

$$= \int_{0.5}^{1.5} \frac{3}{12} (6u - 3u^2) du$$

$$= \frac{1}{12} \left[\left(\frac{6^3}{3} - \frac{6^2}{2} \right) - \left(\frac{2^3}{3} - \frac{2^2}{2} \right) \right]$$

$$= 4.44 \quad [Am.]$$

$$(iii) \quad E(n^2) = \int_2^6 n^2 \cdot \frac{1}{12} \cdot (n-1) \, dn$$

$$= \frac{1}{12} \int_2^6 (n^3 - n^2) \, dn$$

$$= \frac{1}{12} \left[\frac{n^4}{4} - \frac{n^3}{3} \right]_2^6$$

$$= \frac{1}{12} \left[\left(\frac{6^4}{4} - \frac{6^3}{3} \right) - \left(\frac{2^4}{4} - \frac{2^3}{3} \right) \right]$$

$$= 20.89$$

$$\therefore \text{Var}(n) = E(n^2) - [E(n)]^2$$

$$= 20.89 - (4.44)^2$$

$$= 1.1753$$

$$\therefore SD(n) = \sqrt{\text{Var}(n)}$$

$$= \sqrt{1.1753} = 1.084$$

$$\text{Now, } CV(n) = \frac{SD(n)}{E(n)}$$

$$= \frac{1.084}{4.44}$$

$$= 0.244 \quad [Am.]$$

7. Suppose that in a certain region of a country the daily rainfall is a continuous random variable X with probability density function $f(x)$ given by

$$f(x) = \frac{3}{12} (6x - 3x^2), \quad 0 < x < 2$$

Find the probability that at given day in this region the rainfall is

- (i) not more than 1.5 inches
- (ii) between 0.5 and 1.5 inches
- (iii) also calculate mean and variance of the daily rainfall.

Solⁿ:- Given that,

$$f(n) = k(n-1) ; 2 \leq n \leq 6$$

(i) we know,

$$\int_2^6 f(n) dn = 1$$

$$\Rightarrow \int_2^6 k(n-1) dn = 1$$

$$\Rightarrow k \left[\frac{n^2}{2} + n \right]_2^6 = 1$$

$$\Rightarrow k \left[\left(\frac{6^2}{2} + 6 \right) - \left(\frac{2^2}{2} + 2 \right) \right] = 1$$

$$\Rightarrow k \cdot \frac{24}{2} = 1$$

$$\therefore k = \frac{1}{12} \quad [Ans.]$$

$$(ii) E(n) = \int_2^6 n f(n) dn$$

$$= \int_2^6 n \cdot \frac{1}{12} \cdot (n-1) dn$$

$$= \frac{1}{12} \int_2^6 (n^2 - n) dn$$

$$= \frac{1}{12} \left[\frac{n^3}{3} - \frac{n^2}{2} \right]_2^6$$

$$= \frac{1}{4} \left[3x^2 - x^3 \right]_{0.5}^{1.5}$$

$$= \frac{1}{4} \left[3(1.5)^2 - (1.5)^3 \right] - \left[3(0.5)^2 - (0.5)^3 \right]$$

$$= 0.6875 \quad [Am.]$$

$$(iii) \text{ Mean, } E(x) = \int_0^2 x f(x) dx$$

$$= \int_0^2 x \cdot \frac{3}{12} (6x - 3x^2) dx$$

$$= \frac{1}{4} \int_0^2 (6x^2 - 3x^3) dx$$

$$= \frac{1}{4} \left[2x^3 - \frac{3x^4}{4} \right]_0^2$$

$$= \frac{1}{4} \left(2 \times 2^3 - \frac{3 \times 2^4}{4} \right)$$

$$= \frac{1}{4} \times 4 = 1 \quad [Am.]$$

$$\text{Now, } E(x^2) = \int_0^2 x^2 f(x) dx$$

$$= \int_0^2 x^2 \cdot \frac{3}{12} (6x - 3x^2) dx$$

$$= \frac{1}{4} \int_0^2 (6u^3 - 3u^4) du$$

$$= \frac{1}{4} \left[\frac{6u^4}{4} - \frac{3u^5}{5} \right]_0^2$$

$$= \frac{1}{4} \cdot \left(\frac{6 \times 2^4}{4} - \frac{3 \cdot 2^5}{5} \right)$$

$$= 1.2$$

$$\therefore \text{Var}(u) = E(u^2) - [E(u)]^2$$

$$= 1.2 - 1 = 0.2 \quad [Ans.]$$

8. A software engineer records an average of 5 login failures per hour on the university portal. Find the probability that -

- (i) Exactly 4 login failure occur.
- (ii) At most 2 login failure occur.
- (iii) At least 3 login failure occur.

Solⁿ: Given that,

Average number of login failure per hour, $\lambda = 5$

Let, x be the number of login failure per hour.

$$P(x=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

(i) Probability that exactly 4 login failure:

$$\begin{aligned} P(x=4) &= \frac{e^{-5} \cdot 5^4}{4!} \\ &= \frac{e^{-5} \cdot 625}{24} \\ &= 0.1754 \quad [Ans.] \end{aligned}$$

(ii) At most 2 login failure:

$$P(x \leq 2) = P(0) + P(1) + P(2)$$

Here,

$$P(0) = \frac{e^{-5} \cdot 5^0}{0!} = e^{-5} = 0.006737$$

$$P(1) = \frac{e^{-5} \cdot 5^1}{1!} = 0.006737 \times 5 = 0.03369$$

$$P(2) = \frac{e^{-5} \cdot 5^2}{2!} = 0.006737 \times 12.5 = 0.08421$$

$$\therefore P(n \leq 2) = 0.006737 + 0.03369 + 0.08921$$

$$= 0.1246$$

[Am.]

(iii) At least 3 login failure:

$$P(n \geq 3) = 1 - P(n \leq 2)$$

$$= 1 - 0.1246$$

$$= 0.8754$$

[Am.]

Q. In a software development company a project manager is interested in investigating whether there is a correlation between the type of programming language used (python or Java) and the time of the day when coding errors occur (morning or

evening). This project manager collects data on coding errors over a month and constructs a 2×2 contingency table as follow:-

	Python	Java
Morning	40	15
Evening	25	30

Conduct a chi-square test at a significance level of 5% of determine whether there is an association between the programming language used and the the time of day when coding errors occur.

Solⁿ : Observed frequency table :

	Python	Java	Row total
Morning	40	15	55
Evening	25	30	55
Column Total	65	45	110

By using the formula :

$$E_{ij} = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

Now,

$$\text{Morning - python} : \frac{55 \times 65}{110} = 32.5$$

$$\text{Morning - Java} : \frac{55 \times 45}{110} = 22.5$$

$$\text{Evening - python} : \frac{55 \times 65}{110} = 32.5$$

$$\text{Evening - Java} : \frac{55 \times 45}{110} = 22.5$$

Compute the chi-square statistic:

$$\chi^2 = \frac{(40-32.5)^2}{32.5} + \frac{(15-22.5)^2}{22.5} + \frac{(25-32.5)^2}{32.5} + \frac{(30-22.5)^2}{22.5}$$

$$= 1.73 + 2.5 + 1.73 + 2.5$$

$$= 8.46$$

[Ans.]

————— 0 —————